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MCN201 SUSTAINABLE ENGINEERING

MODULE 3

“Environmental management standards: ISO 14001:2015 frame work and benefits, Scope and goal of Life Cycle Analysis (LCA), Circular economy, Bio-mimicking, Environment Impact Assessment (EIA), Industrial ecology and industrial symbiosis.”

3.1 ENVIRONMENTAL MANAGEMENT STANDARDS:

An Environmental Management System (EMS) is a set of management procedures that allows an organization to **identify, evaluate and reduce the environmental impact** of its activities. An EMS can also help address non-regulated issues such as **pollution prevention, odor management, and water and energy conservation**. In addition, an EMS can promote stronger operational control, and as a result reduce inefficiency and waste.

EMS Benefits:

- Improved overall environmental performance
- Identification of pollution prevention opportunities
- Enhanced operational control and efficiency
- Cost savings
- Reduced risk (environmental, health and safety)
- Improved internal communication
- Improved external relations and public image

3.2 The ISO14000 Standards

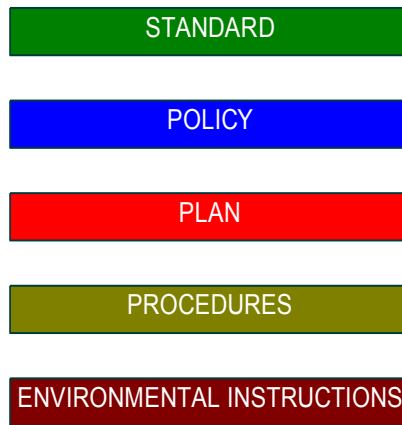
Introduced in 1996, the ISO 14000 series of standards are internationally recognized standards for environmental management. The ISO14000 series includes standards for Environmental management systems,

- Environmental auditing,
- Environmental performance evaluation,
- Environmental labels and declarations and
- Life cycle assessment.

3.2.1 BENEFITS OF ISO 14000

- Helps organizations meet and maintain regulatory and legislative requirements
- Minimizes the environmental impact of products, activities, and resources.
- Promotes environmental awareness

3.2.2 DOCUMENTATION STRUCTURE OF ISO 14000



3.2.3 TARGETS & OBJECTIVES OF ISO 14000

- Reduce waste generation by recycling
- Reduce energy consumption by use of alternative lighting
- Reduce energy consumption by reduction of compressed air leaks
- Improve chemical management system software
- Improve management of Industrial Wastewater
- Improve waste management through solvent recovery

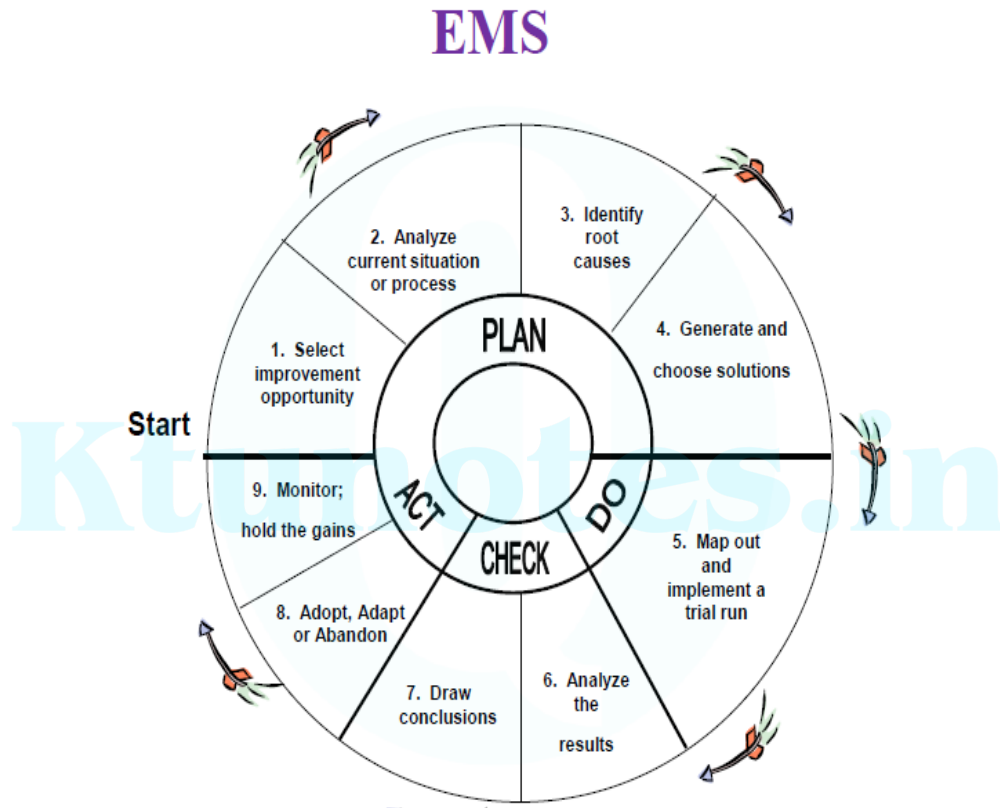
3.2.4 BASIC ELEMENTS OF AN EMS:

- **Reviewing** the company's environmental goals
- **Analyzing its environmental impacts** and legal requirements
- **Setting environmental objectives** and targets to reduce environmental impacts
- **Establishing programs** to meet these objectives and targets
- **Monitoring and measuring** progress in achieving the objectives
- Ensuring employees' environmental **awareness** and competence
- Reviewing progress of the EMS and making improvements

EMS mainly serves the following functions:

- Serves as a tool to provide a systematic way of managing an organization's environmental affairs
- Gives order & consistency for organizations to address environmental concerns
- Focuses on continual improvement of the system

The EMS framework follows a **PLAN-DO-CHECK-ACT** (PDCA) cycle



PLAN -Planning, identifying environmental aspects & establishing goals in accordance with the organization's environmental policy

DO -Implement the planned processes which includes training & operational controls

CHECK -Checking (monitoring) & corrective actions

ACT -Reviewing, includes progress reviews & actions to make needed changes which continually improve performance of the environmental management system

3.3 LIFE-CYCLE ANALYSIS/ASSESSMENT (LCA)

Life-cycle analysis (LCA) is a method in which the **energy and raw material consumption, different types of emissions** and other important factors related to a specific product **are being measured**, analyzed and summoned over the products entire life cycle from an environmental point of view. LCAs started in the early 1970s, initially to investigate the energy requirements for different processes. Emissions and raw materials were added later. LCAs are considered to be the most comprehensive approach to assessing environmental impact

Nowadays, ISO 14040-14043 is considered to be the LCA standard

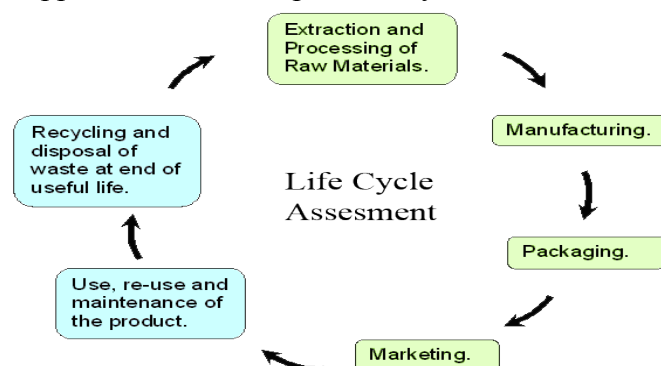
Goals and purpose

The goal of LCA is to compare the full range of environmental effects assignable to products and services by quantifying all inputs and outputs of material flows and assessing how these material flows impact the environment. This information is used to improve processes, support policy and provide a sound basis for informed decisions.

“The term life cycle refers to the assessment of raw-material production, manufacture, distribution, use and disposal including all intervening transportation steps necessary or caused by the product's existence.”

There are two main types of LCA.

- **Attributional LCAs** seek to establish the burdens associated with the production and use of a product, or with a specific service or process, at a point in time (typically the recent past).
- **Consequential LCAs** seek to identify the environmental consequences of a decision or a proposed change in a system under study (oriented to the future), which means that market and economic implications of a decision may have to be taken into account.
- **Social LCA** is under development as a different approach to life cycle thinking intended to assess social implications or potential impacts. Social LCA should be considered as an approach that is complementary to environmental LCA.



3.3.1 Generally, a LCA consists of four main activities:

1. Goal definition (ISO 14040):

The basis and scope of the evaluation are defined.

2. Inventory Analysis (ISO 14041):

Create a process tree in which all processes from raw material extraction through waste water treatment are mapped out and connected and mass and energy balances are closed (all emissions and consumptions are accounted for).

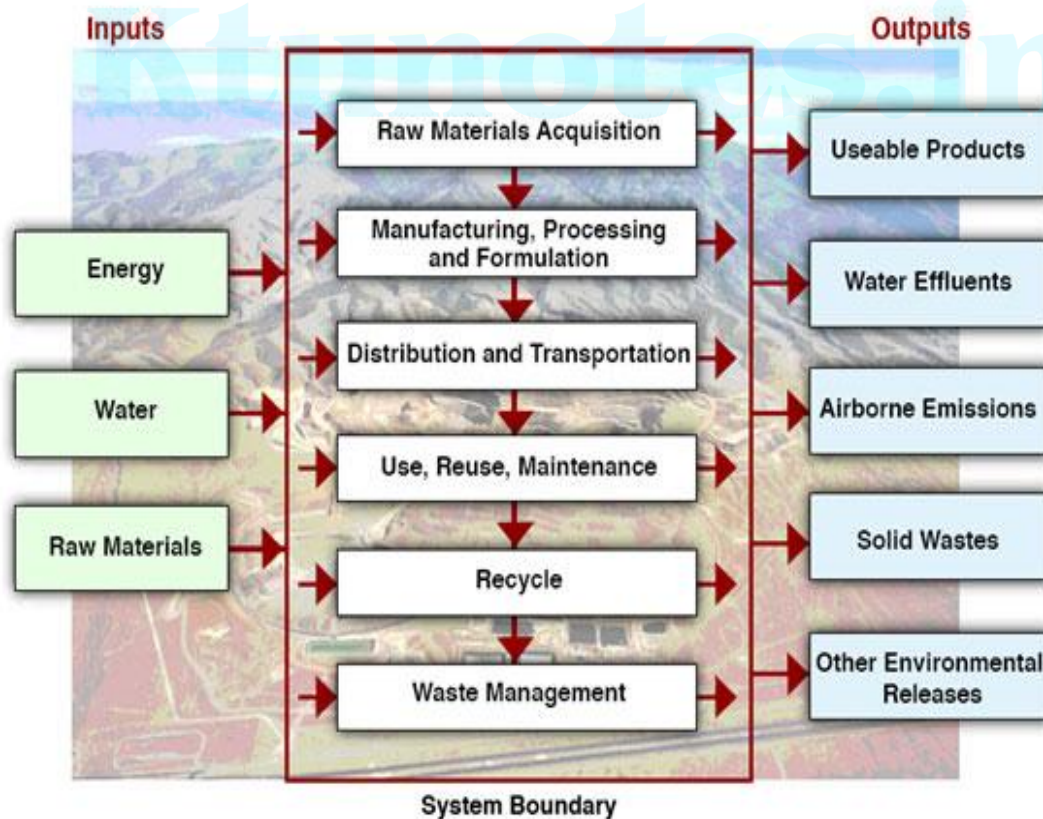
3. Impact Assessment (ISO 14042):

Emissions and consumptions are translated into environmental effects. The are environmental effects are grouped and weighted.

4. Improvement Assessment/Interpretation (ISO 14043):

Areas for improvement are identified.

LCA OVER VIEW



3.3.2 LCA IN WASTE MANAGEMENT

LCA has begun to be used to evaluate a city or region's future waste management options. The LCA, or environmental assessment, covers the environmental and resource impacts of alternative disposal processes, as well as those other processes which are affected by disposal strategies such as different types of collection schemes for recyclables, changed transport patterns and so on.

The complexity of the task, and the number of assumptions which must be made, is shown by the simplified diagram (above) showing some of the different routes which waste might take, and some of the environmental impacts incurred along the way. Those shown are far from exhaustive.

WHY PERFORM LCAS?

LCAs might be conducted by an industry sector to enable it to identify areas where improvements can be made, in environmental terms. Alternatively the LCA may be intended to provide environmental data for the public or for government. In recent years, a number of major companies have cited LCAs in their marketing and advertising, to support claims that their products are 'environmentally friendly' or even 'environmentally superior' to those of their rivals. Many of these claims have been successfully challenged by environmental groups.

All products have some impact on the environment. Since some products use more resources, cause more pollution or generate more waste than others, the aim is to identify those which are most harmful.

Even for those products whose environmental burdens are relatively low, the LCA should help to identify those stages in production processes and in use which cause or have the potential to cause pollution, and those which have a heavy material or energy demand. Breaking down the manufacturing process into such fine detail can also be an aid to identifying the use of scarce resources, showing where a more sustainable product could be substituted.

3.4 BIOMIMICKING

It is the imitation of the models, systems, and elements of nature for the purpose of solving complex human problems. Living organisms have evolved well-adapted structures and materials over geological time through natural selection. Biomimetic has given rise to new technologies inspired by biological solutions at macro and Nano scales. Humans have looked at nature for answers to problems throughout our existence. Nature has solved engineering problems such as self-healing abilities, environmental exposure tolerance and resistance, hydrophobicity, self-assembly, and harnessing solar energy.

Some applications

- Aircraft wing design[4] and flight techniques[5] inspired by birds and bats
- Robots based on the physiology and methods of locomotion of animals
- Nanotechnology surfaces that recreate properties of shark skin
- Treads on tires inspired by the toe pads of tree frogs
- Self-sharpening teeth found on many animals, copied to make better cutting tools
- Polar bear fur inspired thermal collectors and clothing
- Mimicking the arrangement of leaves on a plant for better solar power collection



Transportation: How does
Nature travel quickly and
smoothly?

The Bullet Train in Japan



3.5 ENVIRONMENTAL IMPACT ASSESSMENT

It is the process used to predict the environmental consequences (positive or negative) of a plan, policy, program, or project prior to the decision to move forward with the proposed action.

It is also defined as “the process of identifying, predicting, evaluating and mitigating the biophysical, social, and other relevant effects of development proposals prior to major decisions being taken and commitments made.” Environmental Impact Assessment (EIA) is an important management tool for ensuring optimal use of natural resources for sustainable development.

3.5.1 THE STAGES OF ENVIRONMENTAL IMPACT ASSESSMENT

There are five broad stages to the process:

➤ **SCREENING**

Determining whether a proposed project falls within the remit of the Regulations, whether it is likely to have a significant effect on the environment and therefore requires an assessment.

➤ **SCOPING**

Determining the extent of issues to be considered in the assessment and reported in the Environmental Statement. The applicant can ask the local planning authority for their opinion on what information needs to be included (which is called a ‘scoping opinion’).

➤ **PREPARING AN ENVIRONMENTAL STATEMENT**

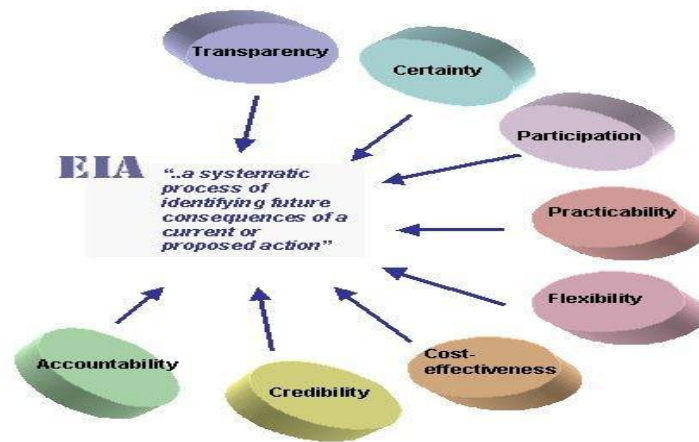
Where it is decided that an assessment is required, the applicant must compile the information reasonably required to assess the likely significant environmental effects of the development. To help the applicant, public authorities must make available any relevant environmental information in their possession. The information finally compiled by the applicant is known as an Environmental Statement.

➤ **MAKING A PLANNING APPLICATION AND CONSULTATION**

The Environmental Statement (and the application for development to which it relates) must be publicized. The statutory ‘Consultation Bodies’ and the public must be given an opportunity to give their views about the proposed development and the Environmental Statement.

➤ **DECISION MAKING**

The Environmental Statement, together with any other information which is relevant to the decision, comments and representations made on it, must be taken into account by the local planning authority and/or the Secretary of State in deciding whether or not to give consent for the development. The public must be informed of the decision and the main reasons for it.



3.5.2 TYPES OF IMPACTS

- Direct & indirect impacts
- Short-term & long-term impacts
- Adverse & beneficial impacts
- Cumulative impacts

3.6 INDUSTRIAL ECOLOGY (IE)

IE is the study of material and energy flows through industrial systems. Industrial ecology conceptualises industry as a man-made ecosystem that operates in a similar way to natural ecosystems, where the waste or by product of one process is used as an input into another process. Industrial ecology interacts with natural ecosystems and attempts to move from a linear to cyclical or closed loop system. Like natural ecosystems, industrial ecology is in a continual state of flux.

Industrial ecology tries to analyse and optimize the following:

- The utilization of waste and by-products in a useful manner
- analyse the losses and hazardous effects while consuming the products, at the time of manufacturing.
- The flow of matter and energy at all its stages must be the minimum.
- Industry should depend more on renewable sources of energy

Within the industrial ecology concept, industry interacts with nature and utilises the wastes and by-products of other industries as inputs into its own processes. Industrial ecology ranges from purely industrial ecosystems to purely natural ecosystems with a range of hybrid industrial/natural ecosystems in between. Covering industrial management and technology, industrial ecology encompasses other sustainability concepts and tools such as material flows analysis; environmentally sound technologies; design for disassembly; and dematerialisation

3.7 INDUSTRIAL SYMBIOSIS

It is the **sharing of services, utility, and by-product resources** among industries in order to add value, reduce costs and improve the environment. Industrial symbiosis is a subset of industrial ecology, with a particular focus on material and energy exchange.

The word “symbiosis” is usually associated with relationships in nature, where two or more species exchange materials, energy, or information in a mutually beneficial manner. Local or wider co-operation in industrial symbiosis can reduce the need for virgin raw material and waste disposal, thereby closing the material loop – a fundamental feature of the circular economy and a driver for green growth and eco-innovative solutions. It can also reduce emissions and energy use and create new revenue streams.

ADVANTAGES OF INDUSTRIAL SYMBIOSIS

- Reduce raw material and waste disposal costs
- Generate new revenue from residues and by-products
- Divert waste from landfill and reduce carbon emissions
- Open up new business opportunities
- Strengthen environmental profiles

